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# TECHNOLOGY ENTREPRENEURSHIP AND INNOVATION DEVELOPMENT IN NIGERIA: A STUDY OF INNOVATION HUBS IN LAGOS STATE

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## ABSTRACT

Technology Entrepreneurship has been identified as one of the core entrepreneurship endeavour that has very high potential of improving the quality of life and economy of any nation, when it is well cultivated and applied to key gap areas within such economy. Incubation hubs as a major centre for the cultivation of such skill and enterprise play major roles in the technology entrepreneurship ecosystem and as such, this research studied forty (40) incubation hubs in Lagos State through the administration of open-ended questionnaire administered on the users of the hubs and face-to-face interviews with the initiators and founders of the hubs. The study discovered that all the technology innovation hubs provided office space with internet and the serenity required to grow entrepreneurship ideas but majority of them lack the conscious entrepreneurship development process that takes cognisance of the relevance of the ideas and the steps required to crystalize such ideas into viable products and services that will generate the real value. About 15% of the sample hubs that are well bred and equipped has created other organizations and incubation hubs and has also attracted funding from various forms of investors, seed funders and corporate organizations to grow and nurture the ideas of their users, some of which has developed products of international patronage. This study re-affirms that the provision of workspace, constant power supply and internet access are fundamentals to growing business ideas but are far from the real entrepreneurship structure where idea discovery, nurturing, funding and commercialization can be developed. In this vein, it is recommended that incubation hubs operators should not just be mere businessmen but passionate individuals and organizations who are strategic enough to create platforms and systems where complete technology entrepreneurship process should take place and thrive.

**Keywords:** Entrepreneurship, Innovation Hubs, Lagos State, Technology Entrepreneurship

## 1. INTRODUCTION AND BACKGROUND

Entrepreneurship and innovation drive economic growth; it helps to create and recreate capacities and wealth; all nations of the world that have embraced entrepreneurship has validated its relevance by the results they have garnered over the years (Akanle & Abraham, 2017; Ikebuaku, 2016).. In Nigeria, several efforts have been made by individuals, corporate organisations, State and Federal tiers of Government to improve the entrepreneurship drive of the Nigerian people, this study looks at that alongside selected incubation centres and their activities in Lagos State, Nigeria. In this very passionate, highly inquisitive, and challenging generation, strategic technology entrepreneurship is highly required to bring about creativity in product design and service delivery for national and global development Technology and innovation have helped entrepreneurship development and exploits so immensely, and the entrepreneurs of the future now build the skills in their very early years.

### a) *Statement of the Problem*

The problem of poverty, unemployment, insecurity etc. are major problems confronting almost all developing countries; in fact, the level of self-sustenance and wealth of any country is one of the major factors determining how developed a country or economy is. Products, developed through entrepreneurial activities are one of the major breakthroughs a nation can have in solving its problem of hunger, unemployment, insecurity etc. (Ikebuaku, 2016); Innovation and entrepreneurship are two inter-related concepts and endeavor that has be proven to help solve these problems when applied strategically. Innovation hubs are training and empowerment centres for equipping current and future entrepreneurs, for networking and for strategizing. The intentional drive of all stakeholders will help in ensuring the various innovation hubs spring up to be more productive and will also help to create specialised hubs focusing on solving problems in specific areas of the economy.

### b) *Aim and Objectives of the Study*

To assess the contributions of selected Technology Incubation hubs in Lagos, Nigeria to technology entrepreneurship development. The specific two specific objectives of this study are to:

- i. Evaluate the activities of selected IT Incubation Hubs (IIH) and Technology Innovation Centres (TIC)
- ii. Examine the impact of selected IT Incubation Hubs (IIH) and Technology Innovation Centres (TIC) in terms of job creation and innovative product development.

### c) *Research Questions*

- i. To what extent does technology incubation hub contribute to entrepreneurship Development in Lagos?
- ii. What are the challenges to impactful technology incubation hubs in Lagos?
- iii. How can Technology Incubation hubs be better managed to provide better results?

### d) *Significance of the Study*

The importance of this work is expected to help current and emerging entrepreneurs, funders, investors, and even private and government agencies to approach the issue of technology entrepreneurship from a more conscious and well thought out perspectives to reap more benefits for the various entrepreneurship efforts and endeavours.

### e) *Scope of the Study*

This study is limited to innovation and entrepreneurship activities of technology hubs and innovations centres located in Lagos State, Nigeria, alone excluding those based in institutions of learning. It is adequate in content because of the fact that Lagos State, though small in land mass is one of the most populous state in Africa and the economic hub of Nigeria. Conclusions from this study could be seen to be synonymous with related study in state of similar structure and attributes.

## 2. LITERATURE REVIEW

An entrepreneur is better at taking opportunities, than classrooms examinations. He scans his environment and creatively produces good and/or services that he can exchange for a profit, and that is why the smartest students of marketing, finance and business administration might end up working for an entrepreneur, who might be far less educated.

### Motivation for Entrepreneurship

People starts or operates businesses for different reasons, each of them having different level of push and influence on the actions of the entrepreneurs. Motivation could be seen as the “tipping points” that triggers the individuals to start the process of business creation. Some businesses were started out of necessity while some out of opportunity (Stephan, Hart, Mickiewicz, & Drews, 2015). The report of the research conducted by (Pipirigeanu, Zaman, Strasser, Arama, & Strasser, 2014), (Stephan, Hart, Mickiewicz, & Drews, 2015) posits that autonomy followed by financial motives are the leading motivation for entrepreneurship.

It indicated that seven dimensions as captured below provides entrepreneurial motivation in sufficient breadth and depth as (1) Achievement, challenge and

learning, (2) Independence and autonomy, (3) Income security and financial success, (4) Recognition and status, (5) Family and Roles, (6) Dissatisfaction, and (7) Community and social motivations.

### The Concepts of Creativity

Creativity, which is either adaptive or innovative, is doing things in a different way, it shows a deep understanding of a concept, and that depth of understanding informs the dynamic perspectives at which the concept/object/product/service is viewed and approached. Creativity is borne out of depth both vertically and horizontally, which can be conceptualized by a willing and dynamic mind. It shows genuineness and adaptation to current realities without losing the core focus. Entrepreneurship thrives on creativity because it presents new things/concepts or seemingly old concepts in new ways (Drucker, 1984; KPMG in Nigeria, 2016).

### Innovation and Entrepreneurship

As expressed in Bayarcelik, Tasel, & Apak (2014), SMEs are strategic drivers of innovation and competitiveness in any economy. Entrepreneurship Innovation could be said to be any or combination of (i.) Creation of a new product or service (ii.) Changes in the distribution pattern, properties, packaging etc. of an existing

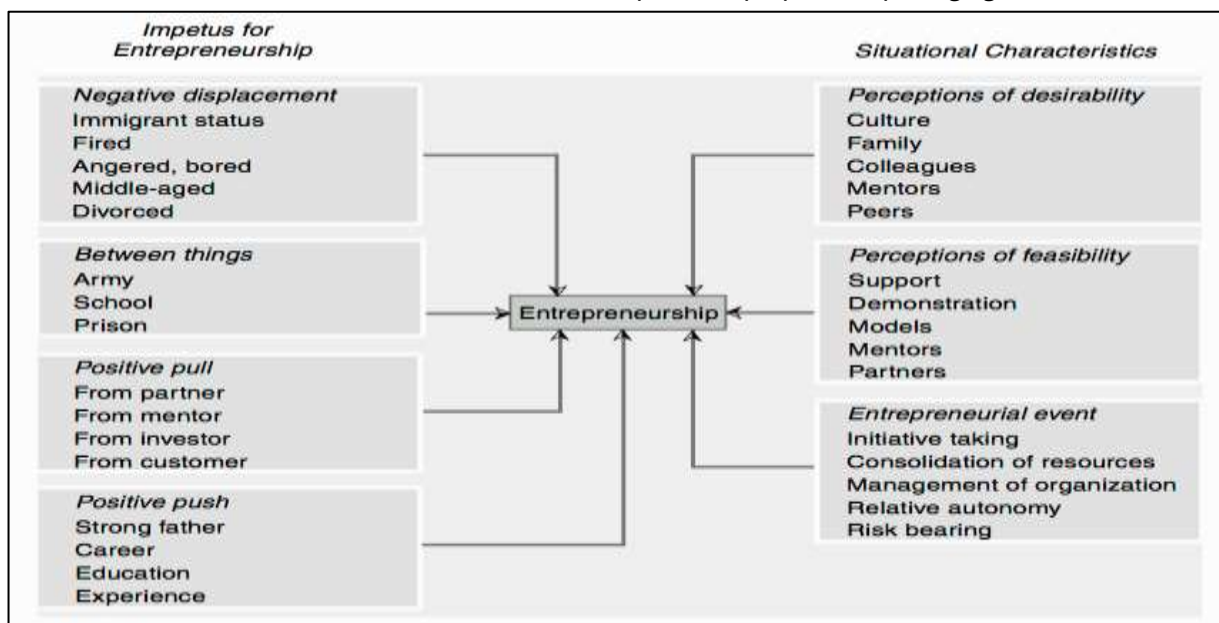
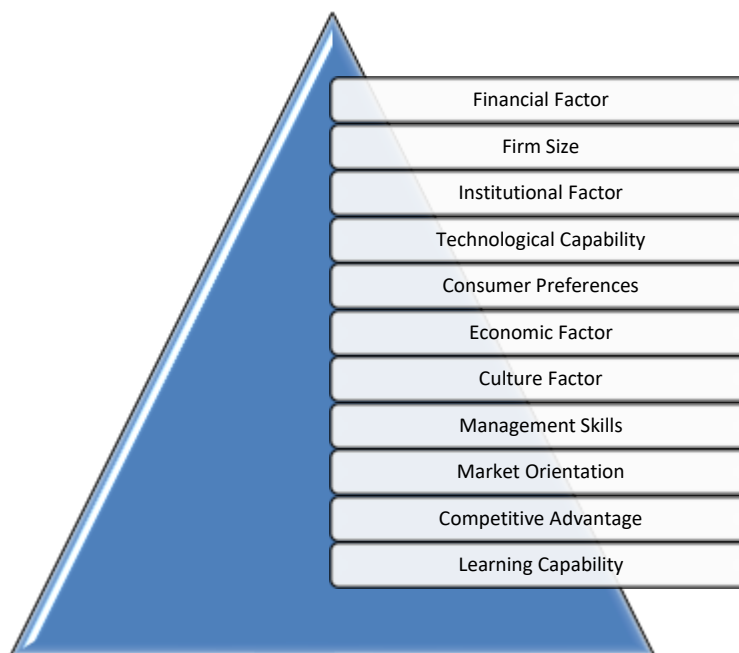


Figure 1: Sociological Approach to Entrepreneurship Source (Dollinger, 2008)

products or service (iii.) Positive shift in the operational strategy of an organisation (vi.) Creation of new markets for new and existing products or/and (v.) Better efficiency in production and product quality (Mehta, Chandani, & Neerajac, 2014). The most

Sanbuli & Whitt (2017) is a research report of the Institute of Development Studies, which emphasizes the need for technology Incubation Centres founders to engage policy makers and funders in a more strategic approach. The authors view a Technology



**Figure 2:** Innovation Factors according to (Bayarcelik, Tasel, & Apak, 2014) as modified

important factors that affects innovation as captured in Bayarcelik, Tasel, & Apak (2014) and reviewed are expressed below:

#### **On technology innovation centres and information technology incubation hubs**

In a 2016 World Development Report titled “How Tech Hubs are Helping to drive Economic Growth in Africa” written by Tim Kelly and Rachel Firestone in (Kelly & Firestone, 2016); the patterns of origin of technology hubs, why they were created, why they have high failure rate and what makes for their successes were reported. The paper tracks about One Hundred and Seventeen (117) technology hubs across Africa. The report was empirical in stating how the rapid spread of digital technology across Africa has not yielded its corresponding dividends. However, no statement was made on whether the spread is commensurate with what is required to have digital independence.

Incubation Centre as a space and community that reduces the barriers to co-creating positive solutions, provides opportunities to build technological skills and competencies in a participatory culture, apply technological know-how and entrepreneurial strength into solving problems, and thinks much more than profitability by including social innovation, community empowerment, and human-centred development.

#### **Funding of Technology and Innovation Driven Development**

Having the right and timely access to funding is a major success factor in any entrepreneurial or innovation driven endeavour, the funding model suitable for different types of innovation driven ideas varies from country to sectors and to object of innovation. Each stage of idea generation to market development and market expansion requires different level and volume of funding.

Considering the high rate of uncertainty that characterizes market acceptance of innovative products and services, the challenge of funding seems more daunting compared to entrepreneurs involved in trading and general merchandise. Apart from the personal savings of its founders and borrowings or grants from the founder's families and friends; other sources of funding opportunities opened to creators and founders of innovative driven ideas are government grants, microfinance banks offering micro credits, angel investors and venture capitals (Bay & Cil, 2016; Beer, Millar, Mwangi, Nzomo, & Rutenberg, 2017; African Union Commission, 2014).

## Some Federal Government Agencies making efforts in Technology Innovation and Entrepreneurship

### I. National Board for Technology Incubation Centre (NBTI) (<https://nbti.gov.ng/>)

The Centre is an agency under the supervision of the federal Ministry of Science and Technology, Nigeria; with the mandate to facilitate the establishment and coordination of the activities of Technology Incubation Centres and Science & Technology Parks, which are designed to nurture new technology, based start-up businesses in Nigeria. Some other specific mandates of the agency are to (i) sensitize all stakeholders for the establishment, supervision and regulation of Technology Incubation Centres, (2) provide institutional infrastructure and mechanisms for the development and commercialization of research and development outputs and inventions, (3) effectively involve all relevant stakeholders in the establishment of Technology Incubation Centres across the country.

The centre is also expected to source and develop commercially viable indigenous and foreign technologies, look for funds for the implementation of the Programme, source for entrepreneurs and investors for the commercialization of chosen technologies. The broad categories of its functions also include ensuring competitiveness of the products of the Programme through value-added service support and effective linkages to knowledge providers

and to ensure continuous injection of innovation in materials processing and/or equipment, ensuring efficient and effective performance leading to fast-tracked growth of entrepreneurs in the Programme. While doing all the aforementioned, the centre is also expected subsidize all services provided by the technology incubation programme in order to reduce the overhead burdens of start-up enterprises and ensure compliance with policy and regulations of the Programme.

NBTI is also charged with the provision of post-incubation survival scheme for the enterprises that graduated from its programme and promote the growth of private sector through the creation of competitive SMEs as the engine of growth of wealth creation, employment generation and poverty reduction. In addition, encouraging value re-orientation to promote hard work, transparency, and accountability in business operations, and interfacing with research establishments and tertiary institutions for the commercialization of their research and developments outputs.

### II. Nigeria Information Technology Development Agency (NITDA) (<https://nitda.gov.ng>)

The agency is under the supervision of the Ministry of Communications, newly renamed as the Ministry of Communications and Digital Economy. The agency has embarked on quite a number of initiatives to promote innovation and technology entrepreneurship, some of which are:

#### 1. Office of ICT Innovation and Entrepreneurship (OIIE), ([https://make-it-initiative.org/africa/cpt\\_partner/oiiel](https://make-it-initiative.org/africa/cpt_partner/oiiel), <https://ictinnovation.gov.ng/our-programs/>)

It has a mandate to nurture, cultivate and expand ICT innovation and entrepreneurship within Nigeria. In recent times, the office has sought to develop the ecosystem in Lagos and Abuja through its flagship "Startup Friday" program aimed at identifying promising startups and providing cash awards.

Some of its other initiatives and programs are policy hackathon programs, technology start ups hints and assistance, mentoring, due diligence training, venture capitalism etc.

2. **Digital Job Creation Initiative**  
(<https://nitda.gov.ng/digital-job-creation-centres/>)

The digital job creation Centre initiative is also a major technology innovation and entrepreneurship development of the National Information Technology Development Agency. The main objective of initiative is to create digital job/wealth, develop digital skills, and promote the connection of underserved areas, schools, libraries and institutions to broadband Internet. The project is aimed at facilitating a digital lifestyle in the institutions, underserved areas (rural), as well as in the communities' schools.

Its other key objectives are to enhance the abilities of communities to access information, manage local level information and disseminate this information to external communities, the creation of enabling environment that will nurture innovation and entrepreneurship, development of training & professional programs to promote self-employment and entrepreneurship and provision, promotion of professional IT skills to community so that they are equipped with industry oriented quality education and training. The initiative is to further introduce practical computer courses to improve computer literacy of students, teachers and parents, impart education and training with the aim of generating quality human resources in the IT industry, and impart a basic level of IT Literacy program for the rural & Semi urban people. Also included in its mandate is the offering of job-oriented

courses like e-commerce, retail management, Web design etc. to enhance IT skill, bridging the gap relative working in neighbouring cities, towns, countries or abroad with those in the rural areas through the Internet and e-mail; and ensuring that local small-scale industries have access to information about market prices thus saving on time and travel costs.

III. **Aso Villa Demo Day 2016**

This was an initiative of the Presidency aimed at developing innovation and technology entrepreneurship in the Nigeria, thirty various startups were invited to pitch at the Presidential Villa (Aso rock) in Abuja after various screening events held in Lagos, Abuja and Port-Harcourt. Before the pitching process, the identified start-ups went through boot camp at Ventures Platform Hub in order to refine their ideas and businesses. At the pitch, the startups had the opportunity to meet potential several financiers and partners as well as pitch to the President, Vice President and Mark Zuckerberg (the CEO of facebook.com) who was also in attendance on invitation.

**Funding model of IT Incubation Hubs (IIH) and Technology Innovation Centres (TIC)**

Funding is an important part of an entrepreneurship endeavour; it provides the glossier and sweetener flavour required to translate ideas into reality. Funding has also become a major source of concern for most entrepreneurs (The African Institution of Technology Inc. (United States of America), 2015) (ASPEN Network of Development Entrepreneurss (ANDE) West Africa Regional Chapter) (Pipirigeanu, Zaman, Strasser, Arama, & Strasser, 2014), (Schwaab, 2017). Various funding methods such as personal savings of the entrepreneurs, his or her family and friends, venture capitalism, angel investors, micro-finance banks, commercial banks, crowd funding, foundations, accelerators/incubators, equity and public offering through various stock exchange

markets are the major options of funding available for entrepreneurs across the globe.

One thing is fund availability, and another is fund accessibility by those who really needs them. Start-up entrepreneurs and budding entrepreneurs will require different volume of funds for different ideas at different stages of their entrepreneurship ventures and as such, some of the funding options available to technology entrepreneurs in Nigeria as chronicled by (Schwaab, 2017) are briefly discussed hereunder:

### **Fastforward Innovation Fund** (<http://fastforward.com.ng>)

This organisation, founded in 2017 in Nigeria is poised to make funds available to technology businesses and ideas by students in tertiary institutions to the tune of One million, four hundred thousand naira.

### **Enspire Hub** (<http://enspire.org.ng>)

It is an incubation centre that makes fund available to its qualified patrons for technology entrepreneurship and innovations to the tune of three hundred and sixty thousand naira only. The hub was founded in Nigeria in 2009.

### **Tony Elumelu Foundation** (<http://tonyelumelifoundation.org>)

A former Managing Director and Chief Executive Officer of United Bank for Africa (UBA), Mr. Tony Elumelu initiate the foundation in 2010 to support various entrepreneurship endeavours across Africa. The foundation provides fund up to the tune of one million, eight hundred thousand naira to qualified entrepreneurs across Africa.

### **Lagos Angel Network** (<http://lagosangelnetwork.net>)

This is an equity funding and networking initiative for entrepreneurs to share ideas and support themselves to the tune of three million, six hundred naira per registered member.

### **Microtraction** (<http://microtraction.com>)

Microtraction is an equity-funding model founded in 2017, in Nigeria for technology-driven initiative, and can give up to twenty-three million naira.

### **Imeela** (<https://imeela.com>)

This was founded in 2016 and with a funding goal centred on elimination of barriers to creativity, innovation, and passion. It supports technology start – ups in Nigeria to the tune of eighteen million naira.

### **Sasware, A signal Alliance Company** (<http://saswarenigeria.com>)

A Nigerian equity-funding initiative that provide funding for sound technology-driven ideas to a maximum limit of ten million naira.

### **Growth Capital by CcHub** (<http://gc.fund>)

It was an African-focused funding model founded in 2016 to support technology initiative in the areas of finance, health, digital security, education, smart infrastructure, and governance. It is a brainchild of the foremost innovation hub in Nigeria, CcHub. It could provide fund up to eighteen million naira to technology driven endeavour in any of its areas of focus.

### **Beta Ventures** (<http://beta.ventures>)

It is a Nigeria based mobile technology equity funding model that supports mobile technology enabled entrepreneurial idea to the tune of thirty – six million naira.

### **Greentree Investment Company** (<http://greentreeinvestmentcompany.com>)

It is a long-term equity/and/or convertible debt-funding model for the finance sector using software as a service. Maximum funding provision of fifty-four million naira is guaranteed with Greentree Investment Company.

### **Research Methodology**

This section presents the techniques and methods of investigation adopted in this study. It justifies why the method is adequate in scope and good enough to bring out the true picture as it relates to the scope and purpose of this study. Various research methods can be adopted for varying levels of research investigations, so, this study combines both

qualitative and quantitative techniques.

### 3. RESEARCH DESIGN

Research methodology is very significant because it bequeaths the systematic ways adopted in responding to the research questions and providing solution to the research problem(s) identified. Research design provides a blueprint on how a specific research endeavor is being carried out; it provides a sequential plan on how the researcher answers the research questions accurately and objectively (Fabiana, Edson, & Marcela, 2016; Mack, Woodsong, MacQueen, Guest, & Namey, 2011). It determines the relevant data and how the data will be collected and analyzed (Bhattacharjee, 2012). Multiple case research by visitation, website reviews; qualitative approach using in-depth interview with initiators of incubation hubs and quantitative approach using survey questionnaires administered on users of incubation hubs were used in this study.

#### Qualitative Research Design

The website of 40 (forty) hubs were visited to get first-hand information on the content of their website as regards their services and offerings, management, goals and motivation, this is adequately reported towards the end of this chapter. In depth interviews were conducted for 29 (twenty –nine) out of the 40 innovation hubs identified in Lagos. The questions bothered on mode of ownership, funding, details of operation and services, motivation, impacts and challenges being faced by these entrepreneurs and trainers. Sample questions used for the interviews are attached as appendix in the work and the findings are reported in chapter 4 of the study.

#### Quantitative Research Design

Quantitative approach was used for the users of the hub; Google form was used as responder's mode of participation while some hard copies were also on hand to be administered to those who feel comfortable using such. The first section of the questionnaire requested for the details of the respondent such as age, sex, academic qualification, years of experience and years of patronage of the

incubation hub. The second section sought to get answers to questions bordering on the entrepreneurial activities of the users of the hub, some the questions border on company registration, years of engagement in entrepreneurial activities, products and services being offered and the sectors of the economy enjoying such services or products. Other questions are focused on funding of their entrepreneurial plans and activities, the type of offerings being enjoyed at the incubation hub and open-ended questions regarding challenges and opportunities. A sample of the quantitative questions in the questionnaire is also enclosed as an appendix in this work.

#### Why use a Combined Research Approach?

Useful insights are provided when qualitative research methods are used together with qualitative techniques, specifically in social and management sciences; it involves several data collection techniques and organized to provide several but not similar data sets regarding the identical phenomena (Guda, Tanase, & Purcarea, 2017; Mack, Woodsong, MacQueen, Guest, & Namey, 2011; Bhattacharjee, 2012).

#### Population of the Study

Lagos, the most densely populated city in Nigeria with a population of over twenty million people and African's 5<sup>th</sup> mega city; have almost 92 innovation hubs (Office of the Vice President, Federal Republic of Nigeria, 2017; KPMG in Nigeria, 2016; Akanle & Abraham, 2017; ASPEN Network of Development Entrepreneurss (ANDE) West Africa Regional Chapter). Lagos, by virtue of its population has its own unique internal mechanisms maintaining interethnic, interreligious and social balances. A foremost seat of federal power as a former capital of Nigeria right from the creation of the state has had its own developmental processes in between different governments and administrations.

Lagos, as an emerging independent economy has embraced model city approaches, welcomed new technologies and increasingly and consistently improving on its fiscal management towards more

growth, funding, private and foreign direct investment. The state is described as one of the most urbanizing state in the world, from a modest population of 25,000 in 1866 to 665,000 in 1966, became over 10 million in 1995 and that makes it a mega city by UN definition, the city now has well over 20 million people as its inhabitants (Okoye, 2016). The total population for the study is about ninety-two (92) and a sample of forty (40) was used. Considering the small number of the population; this excludes the innovation hubs domiciled within academic institutions in the state.

### **Sampling Technique and Sample Size**

Purposeful sampling and quota sampling were used to select the respondents in this work. Purposeful sampling is adequate because of the fewer numbers of the incubation hubs and the specificity nature of their physical location and ownership. Purposeful sampling was used in the selection of the hubs initiator to talk to, few of the incubation hubs numbering about 16 (sixteen) were initiated by individuals while about 10 (ten) were owned and managed by a registered corporate organisation and the remaining were owned by joint ventures including individuals, non-governmental organisations and the organised private sector. In case of multiple ownerships, the choice of who was spoken to was determined by level of stake/ownership and availability. Quota sampling was used in the administration of questionnaire to the patrons/users of the hubs visited. Questionnaires were administered to some of the patrons on ground during the period of physical contact and visitation with the hub; minimum of 2 (two) and maximum of 5 (five) questionnaires were administered at each centre, while some chose to respond to the questionnaire using the online Google form. Considering the small number of incubation hubs in Lagos, numbering about 92, excluding institutions-based technology hubs, a sample of 40 out of the entire population was taken, however only twenty-nine (29) of the initiators were accessible for questioning while questionnaires were administered on users in 33 (thirty-three) of the locations.

### **Data Collection Method**

The period of data collection for this research began immediately the research instrument was approved in February 2020 and was concluded by July of the same year. Aside website, relevant literature and other documents insights, primary data collection through structured in-depth interviews and questionnaire administration was employed as the major means of data collection. The structuring of the questions was systematic enough to provide insights and answers that will lead to giving the correct and sincere answers to the research questions.

### **Data Analysis Method**

Data analysis transforms the research data into useful information, systematically arranged and presented to convey meaning that should lead to further actions (Bhattacharjee, 2012). The recordings for the in-depth interviews were transcribed, properly coded and the salient answers relevant to this work are presented in the chapter four, being conscious of the fact that qualitative research analysis is largely dependent on the investigative, integrative skills and personal knowledge of the researcher. The quantitative data were analyzed using various statistical descriptive methods and Microsoft Excel data analysis tools were employed for this purpose, the details of the quantitative analysis are properly put to shape in chapter four as well.

The table below shows the list of the forty hubs used as sample including their uniform resource locator (url also called website address) and physical contact addresses, where available. Institutions based technology hubs were excluded from this sample.

### **Report of Findings and Discussion**

This section of the article presents the analysis of the qualitative and quantitative data collected from the field as willingly given by the participants/respondents, the analysis seeks to provide insight and the research discovery in answering the research questions and fulfilling the aim and objectives of this study as earlier outlined.

### **Data Presentation and Analysis**

This section presents the analysis in tables, charts and words/expression of the responses received from all

the respondents; first was the analysis of the responses from owners and initiators of technology hubs then the analysis of questionnaire administered on the patrons/users of the technology hubs were presented, analysed and discussed thereafter. Only 6, 15% of the technology hubs have been in existent for over 10 years, while 14, 35% were barely 2 years old.

### Motivation for Establishing the Innovation Hub

From the interview conducted, the motivation for establishing each of the innovation hubs differs one from another, some of the answers from the founders/initiators as the motivation for establishing an incubation hub are as follows in quote:

*"I looked at an average Nigerian youth and put them in my shoes, the struggles I managed alone in my life and the long road I took before gaining some traction and stability in my business; I said to myself let me create a platform for some of these serious boys and let's see if there will be some changes. I started by offering two tables in my former office to four of them to share and from there we are here today having an average of twenty five people per day coming around to work, mingle with others and learn"*

*"It all started right from school, I just thought that I can't cope dropping CV from office to office and we started like a joke, three of us; before we left school, we*

*got a small job and that was more than a push. As boys, after the mandatory national service (NYSC), some of our friends who could not get jobs and who live close by begin to come around, thank God for one of our church member who was an angel to us providing guidance, network and sometime funds for us and today we have two IT services and to create space for our friends and colleagues, this hub started and growing; as you can see we are all young, If government give us funding fine, and if not, we will keep rolling small small"*

*"This is a private sector owned technology hub, it is part of the community social responsibility of the bank to have a technology hub where future techies can be groomed and help contribute their quota to national development"*.

### Services Available at the Hubs

Another important aspect of our question is the type of services being rendered at the hubs; this is displayed for clarity using the figure 4.1. It clearly shows that all of the hubs rendered co-working spaces (mostly paid for by the patrons), 20 of them also have training rooms that is leased or rent out to users and just 5 intentionally have a plan to train and groom the hubs users. All the 5 hubs that have concrete plans and programs to identify, provide platforms and groom the patrons have each developed at least one new product or service since their inception and 2 of the 5

Figure 3: Services Available at the Hubs

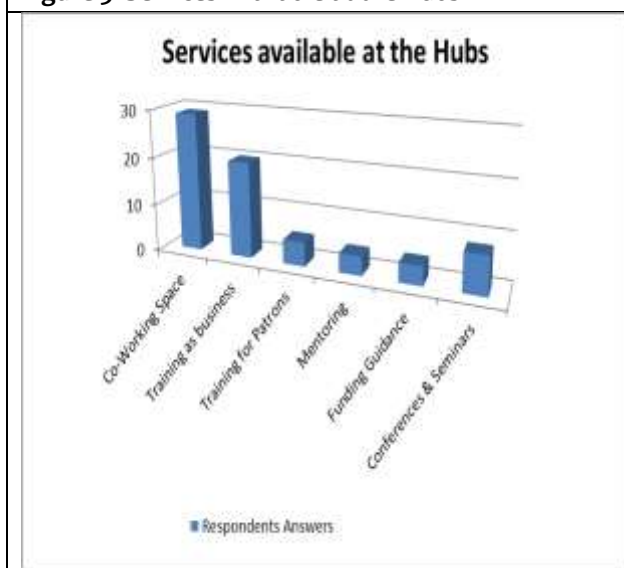
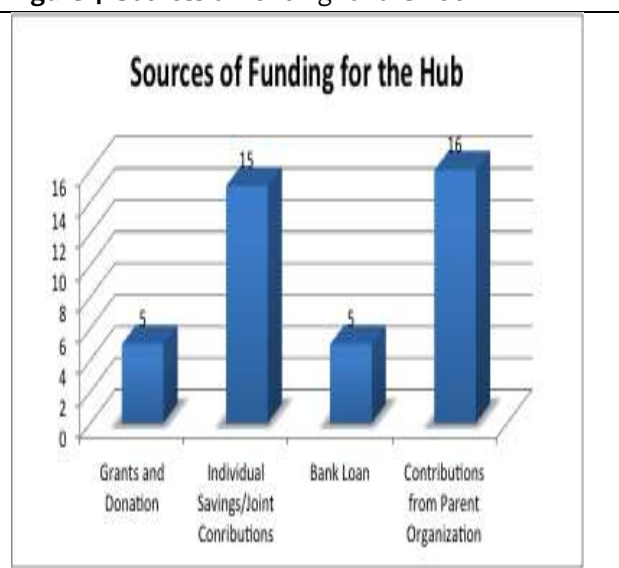


Figure 4: Sources of Funding for the Hub



have launched new businesses out of the products being developed at the hub.

### hallenges of running the Hubs

During the interactive session, questions relating to the challenges encountered as technology hubs owners were asked and their answers identified problems such as power, lack of funding for expansion, lack of exposure and depth by the patrons of the hub. Other problems identified include lack of patience and due diligence on the part of the patrons to enable their idea yield desired results, some of the ideas are not sellable and the process of refining them seems longer because some of the idea owners thinking process are not broad or strong enough. Another major problem identified is the inability of some of the 'boys' readiness to share some part of the business with others.

### Sources of Funding for the Hubs

The research shown that almost all the hubs have more than one funding sources, considering the discovery that corporate entities own majority of them as earlier reported, funding from parent organization was made to 16 of them while individual/joint contributions was one of the funding sources for 15 of the technology hubs, only 5 of them had access to bank loans and another 5 had grants.

### Age range of the Hubs users

From the online and physical questionnaire administered, the highest numbers of users are between ages twenty – six (26) and thirty-five (35) followed by those whose age ranges between thirty-six (36) and forty-five (45) years. Those aged sixty and above patronise the hubs the least, others are as shown in table 4.2 below.

### Length of Hubs Patronage by Users

One of the questions in the questionnaire administered to the hubs users bothered on how long each of them have been using the services of the particular hub where they were met; 47 of them representing 43.12% of those who successfully submitted their questionnaire have been patronising

the hubs for just less than 3 months, while 26, 23.85% have been using that particular hub for between 3 months and 1 year, only a very few fraction of 1.83% have been using the hub for 5 years and above. Let us bear in mind the responses to the length of existence of the hubs that the average age of the hub is barely 3 years. As recorded from the findings none of the patrons had been around in any of the hubs for 10 years because 10 years ago the technology incubation hubs were non-existing within the Nigeria space.

### Hubs Users with at least a registered business

54% of the hubs' users have no registered business yet while the remaining 53, 46% has at least a registered business (Sole Proprietorship or Limited Liability Company).

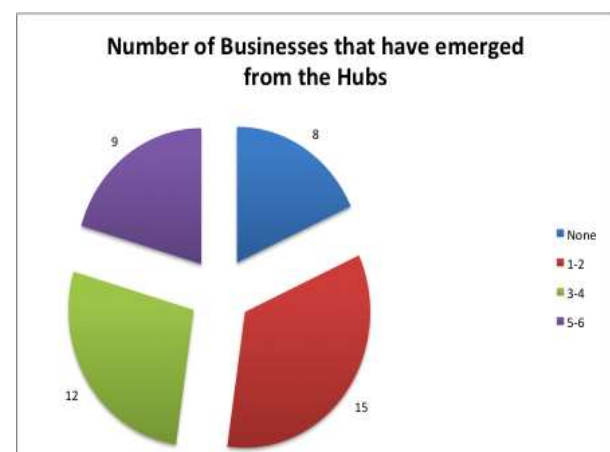


Figure 5: Number of businesses that have emerged from the Hubs

Table 5: Length of Hubs Patronage by Users

| Length of Hub Patronage     | Number |
|-----------------------------|--------|
| Less than 3 months          | 47     |
| Between 3 months and 1 year | 26     |
| Between 1 and 2 years       | 19     |
| Between 3 and 5 years       | 15     |
| Between 5 and 10 years      | 2      |
| Above 10 years              | 0      |

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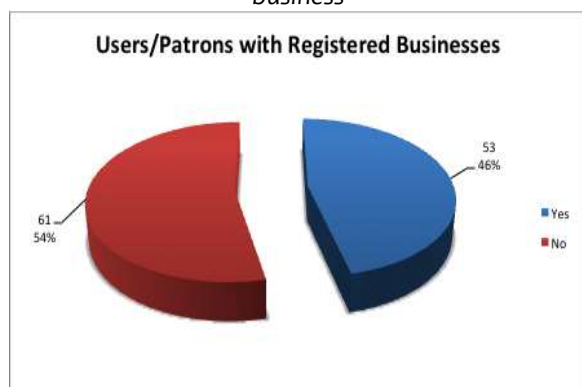
### Length of Hubs Patronage by Users

One of the questions in the questionnaire administered to the hubs users bothered on how long each of them have been using the services of the particular hub where they were met; 47 of them representing 43.12% of those who successfully submitted their questionnaire have been patronising the hubs for just less than 3 months, while 26, 23.85% have been using that particular hub for between 3 months and 1 year, only a very few fraction of 1.83% have been using the hub for 5 years and above. Let us bear in mind the responses to the length of existence of the hubs that the average age of the hub is barely 3 years. As recorded from the findings none of the patrons had been around in any of the hubs for 10 years because 10 years ago the technology incubation hubs were non-existing within the Nigeria space.

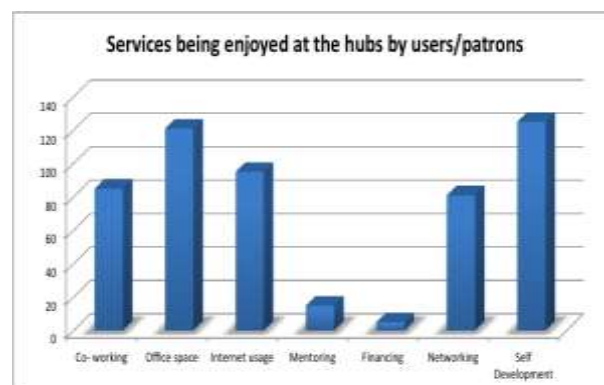
### Hubs Users with at least a registered business

54% of the hubs' users have no registered business yet while the remaining 53, 46% has at least a registered business (Sole Proprietorship or Limited Liability Company).

**Figure 6: Users of Hubs with at least a registered business**



**Figure 8: Services being enjoyed at the hubs by the users**



### Services being enjoyed at the hubs by the users

The core for the establishment of a technology hub is the value being derived from the services being rendered. Figure 4.7 shows the various services being enjoyed by the users of the hubs. This same questioned was asked the creators/founders of the hubs, the essence of this is to cross check and validate the value exchange from both the providers and consumers of the services, the full discussion of this is contained in the section discussing the research findings.

All the incubation hubs provide office spaces for the patrons to use for their activities, other services are as provided on figure 4.7 below.

### Expected sources of funding for the business ideas

As corroborated by Carayannis E, (2013) and Drucker (1984), the importance of funding in translating an idea to a business venture is a key aspect of the entrepreneurial process and this necessitate the question on the likely funding sources being thought of or planned towards by the users of the hubs. to oil their ideas into fruition. The highest responses of 45, 26.63% cannot fathom any idea of where the funds will be coming from as at the time of filling the questionnaire while 30, 17.75% said their personal savings is one of their funding sources, 25, 14.79% planned on getting funds from their various families

and friends while none of the respondents intends to use a commercial bank or public offer. The detail responses to the question are as contained in table 4.8.

**Table 7:** Expected sources of funding for the business idea

| Expected Sources of funding for the business idea |        |
|---------------------------------------------------|--------|
| Sources                                           | Number |
| I do not know                                     | 45     |
| Personal Savings                                  | 30     |
| Family and/or friends                             | 25     |
| Venture Capitalist                                | 10     |
| Angel Investor                                    | 8      |
| Micro finance bank                                | 19     |
| Crowd funding                                     | 12     |
| Equity                                            | 20     |
| Public Offer                                      | 0      |
| Commercial bank                                   | 0      |
| Others, Please Specify                            | 0      |

## Discussion of Findings

This section provides robust discussions on the findings of this research as it relates to answering the research questions identified.

### Contribution of technology incubation hub to entrepreneurship development in Lagos

As reported in the findings in section 4.2, more than 90% of the incubation hubs provide office space and/or co-working space to their users/patrons. This provision provides conducive thinking and working environment for ideas to be developed and nurtured, to this extent the incubation hubs help to contribute to entrepreneurship development. The adequacy of their spread or otherwise to cater for the population of *Lagosians* is outside the scope of this study except for few of the reviewed literatures that made some assertions on their adequate spread. Responses regarding the type of services enjoyed at the hub were recorded from both users and initiators of the hubs, the responses corroborated each other; in this corroboration, it was crystal clear that quite a number

of conscious services that will help accelerate the development of entrepreneurial activities were not offered by not less than 80% of the technology hubs; some of such services include, mentoring, guidance on getting funds, training etc. From the responses relating to sources of funding for the hub, only 10 out of 41 responses shows a hybrid of loan or/and grant as sources for funding the hub, this further clarify the fact that more than 80% of the hubs owners were not successful at raising capital themselves and so, could not have assisted upcoming entrepreneur to raise fund outside personal savings and loans from family and friends.

### Contribution of technology innovation hubs to job creation in Lagos

Job creation is one of the cardinal benefits of entrepreneurship. Every entrepreneur has created an engagement system for himself or herself, thereby moving a step higher from being an employee to first being self-employed and later on providing employment opportunities to others. Findings from this study shows an average of more than 500 users patronising the services of these technology hubs in a day; this shows capacity, knowing fully well the importance of one entrepreneur who got it right to his/her community. Every new business or expanded business creates more employment opportunities for both skilled and semi-skilled workers as well; depending on the nature of products or services the business operates. This study shows that innovation hubs grooms citizens to create employment opportunity, whether the number is adequate compared to the population and expectations of people is beyond the scope of this study.

### Challenges to impactful technology hubs operations in Lagos

From the findings earlier presented in this study, it is reported that the technology incubation hubs were not operating optimally, this is as a result of several factors such as (i) the knowledge of most of the hubs owners and operators are not adequate in scope to provide high level support to budding entrepreneurs that patronizes their hubs, (ii) some of the entrepreneurs do not have systems thinking and are

adamant in growing up, this limits the type of products and services they can come up with and the extent of growth to the very few ones, (iii) the available funding options to starters are very limited and most time cumbersome to access (iv) there are more individuals working alone instead of working with others and leveraging on the skills, experiences and expertise of each other to create relevant products and/or services. These problems are apart from the regular ones facing busy state in a growing economy like Lagos, Nigeria such as power, time it requires to commute from one location to another, too expensive internet access, lack of adequate support from government amongst others. When these problems are looked at, then conscious efforts will ensure the availability of services such as trainings for users, access to markets, access to funds, business advisory, product development guide, research etc. in Lagos technology incubation hubs.

#### 4. RECOMMENDATIONS

The number of incubation hubs and the geographical spread compared with the population of the budding youth resident in Lagos State is grossly inadequate if our journey to wealth creation and entrepreneurship would be worth its while. Entrepreneurship development should be a conscious effort and the driver should understand its clear difference from mere trading.

This work has brought to the forefront again, the engagement of technology hubs and their contribution to wealth creation and national development, a hybrid of research approach adopted to engage both patrons and owners of the incubation hubs has exposed some of the areas that needs to be improved upon, which when worked on will translate to a more prosperous society.

The funding model for existing incubation hubs is more of a personal endeavour at the very basic level of entrepreneurship development, this is slowing down the pace of training and enhancements of entrepreneurship activities compared with the demands and population of Lagos and by extension, the Nigerian populace. Therefore proper financing systems is advised to be set up for owners and

initiators of hubs to access and by extension lead upcoming entrepreneurs on how to access funds both locally and internationally in funding their ideas. Entrepreneurship development should be pursued with more rigour with more enduring strategic plans and processes by government, private sectors, non-governmental organizations, and willing individuals at all levels. Nigeria, with over 65% of its over 200 million citizens populations are less than 25 years old (Office of the Vice President, Federal Republic of Nigeria, 2017; KPMG in Nigeria, 2016; Adegbiyi A. P., 2020), this again re-emphasizes the large population of very young Nigeria citizens with all the benefits of youthfulness that should be properly co-ordinated for the creations of varying products and services that can be consumed locally and improved upon, then for exportation; Innovation hubs has enormous capacity for entrepreneurship development serving short, medium and long – term developmental objectives, and so should be harnessed.

Potential and existing passionate visionary hubs initiators and owners should have access to grants and funding that will help to accelerate entrepreneurship development programs and processes. In addition, successful hub operators who had flourishing businesses emerging from their hubs should be ready to mentor and guide other technology hubs owners on how to effectively run a hub and make provision for seed funding for both the hubs and the ideas of the patrons.

#### 5. CONCLUSION

From the findings in this study, it is pertinent to note that many efforts are being made by individual businessmen and corporate organizations to establish and operate incubation hubs, which is very commendable and a yearning of citizens, who need the right environment where they can connect, network, and move their entrepreneurship thoughts to product development. However, for the incubation hubs to produce the right type of fruits and the multiplier effects that it is capable of bringing forth, systematic efforts and entrepreneurship development process must be followed, and this will require more of tenacity, conscious efforts of record keeping and

follow up, training and re-training of incubators initiators and owners.

### **Suggestion for Further Studies**

This study is limited by scope to incubation hubs outside the shores of any higher institutions of learning in Nigeria, further studies to provide more robust and comprehensive reports on the activities of incubation hubs should include government-owned incubation hubs (at different tiers of government) and those incubation hubs within the universities and polytechnics environment. This will provide a more robust report and guide to have recommendations that will be far reaching and engaging at the national level. Further studies could also include the products being developed by users of these hubs and the relevant of such products to both local and international demands.

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